

RAMESH RANGARAO PATIL

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Professional Summary

Software Engineer specializing in Applied AI and Machine Learning, with 2+ years of experience building scalable backend systems and production-ready AI platforms. Skilled in Python, FastAPI, PyTorch, Generative AI, Agentic Workflows, and RAG architectures, with expertise in microservices, distributed systems, vector databases, ML inference pipelines, and cloud-native infrastructure. Proven ability to deliver intelligent and scalable solutions that bridge advanced AI capabilities with real-world engineering challenges.

Experience

Dynamisch IT Pvt. Ltd.

Software Engineer (AI/ML)

Jan 2024 – May 2026

Pune, Maharashtra

Event-Driven AI Orchestration & Graph RAG Pipeline — Investigation App

- Designed a distributed, event-driven pipeline (FastAPI, RabbitMQ, Celery) to ingest multi-modal files (video, audio, documents) and orchestrate multiple ML models to process them into structured Markdown artifacts.
- Built end-to-end ingestion into a knowledge graph with Graph RAG-based retrieval; implemented production observability with latency and error-rate alerting across the pipeline.

Audio Fingerprinting & ML Inference Service — Wazzdat (Microservices Mobile App)

- Built the core ML microservice in FastAPI for real-time audio fingerprinting and music identification; designed, trained, and deployed a custom CNN-based 128-dimensional embedding model, owning the end-to-end ML lifecycle from data preparation and feature engineering to model serving with TensorFlow Serving.
- Optimized large-scale vector similarity search using Milvus (IVF/HNSW indexing) and Redis caching, achieving sub-100ms retrieval latency while reducing inference overhead through request-level caching and intelligent routing strategies.

ML Observability & Automated Retraining Service — Wazzdat

- Built a standalone ML monitoring microservice to detect embedding drift and retrieval quality degradation using Recall@K and accuracy metrics, enabling proactive model performance tracking; automated model retraining and Milvus index retuning triggered by drift signals with zero manual intervention.
- Developed evaluation pipelines to benchmark retrieval accuracy against gold-standard datasets, establishing measurable release criteria before each production rollout.

Concert Events Platform — Wazzdat (Golang Admin & User Services)

- Took end-to-end ownership of the concert events feature: researched affiliate programs, onboarded multi-platform provider APIs, and built Golang admin service modules with cron schedulers for continuous, fault-tolerant data ingestion.
- Implemented user service modules for concert discovery — geospatial filtering, city/genre-wise browsing, nearby events, trending algorithms, ticketing, and user interactions — with Redis caching reducing API response times by 40%.

Personal Projects

ObserveAI – Autonomous AIOps Platform | Python, FastAPI, OpenTelemetry, Docker, LLMs

- Developed an AI-powered observability platform for anomaly detection, root cause analysis, and autonomous incident remediation across distributed microservices, enabling seamless integration and management of applications with minimal human intervention.
- Implemented dynamic service dependency discovery, RAG-based incident knowledge retrieval, and safe Kubernetes or docker remediation workflows through multi-agent AI orchestration.

Geospatial Aerial Land Cover Segmentation | Python, PyTorch, U-Net, Geospatial AI

- Developed a U-Net semantic segmentation model for pixel-level classifying roads, vegetation, urban regions, and water bodies from aerial imagery.

Vehicle Number Plate Recognition | Python, YOLOv8, OCR, PyTorch

- Developed an end-to-end vehicle number plate recognition system using YOLOv8 object detection and OCR-based text extraction.

ATS Scorer and Agentic Career Assistant | Python, LLMs, LangChain, Agno, OCR, RAG

- Developed an agentic AI platform for ATS scoring, resume-job matching, interview preparation, and candidate assessment.

Technical Skills

Languages: Python, Golang, C, C++

ML & AI Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, Scikit-learn, ONNX

LLM & GenAI: ML Inference, RAG Architectures, Prompt Engineering, Agentic Orchestration, Google Gemini API

Backend & Infra: FastAPI, Django, Flask, Microservices, REST APIs, RabbitMQ, Celery, Docker, CI/CD

ML Ops & Observability: ML Monitoring, Drift Detection, Evaluation Pipelines, Automated Retraining, Alerting

Databases & Vector DB: Milvus, Elasticsearch, PostgreSQL, MySQL, MongoDB, Redis

Cloud & Tools: AWS (EC2, S3), NumPy, Pandas, Matplotlib, librosa, Git

Education

SKN Sinhgad Institute of Technology and Science

B.E. Computer Engineering (Honors in Robotics) | CGPA: 8.28

Dr. DY Patil School of Engineering (Polytechnic)

Diploma in Computer Engineering | 77.50%

Aug 2020 – July 2023

Pune, Maharashtra

July 2017 – Oct 2020

Pune, Maharashtra